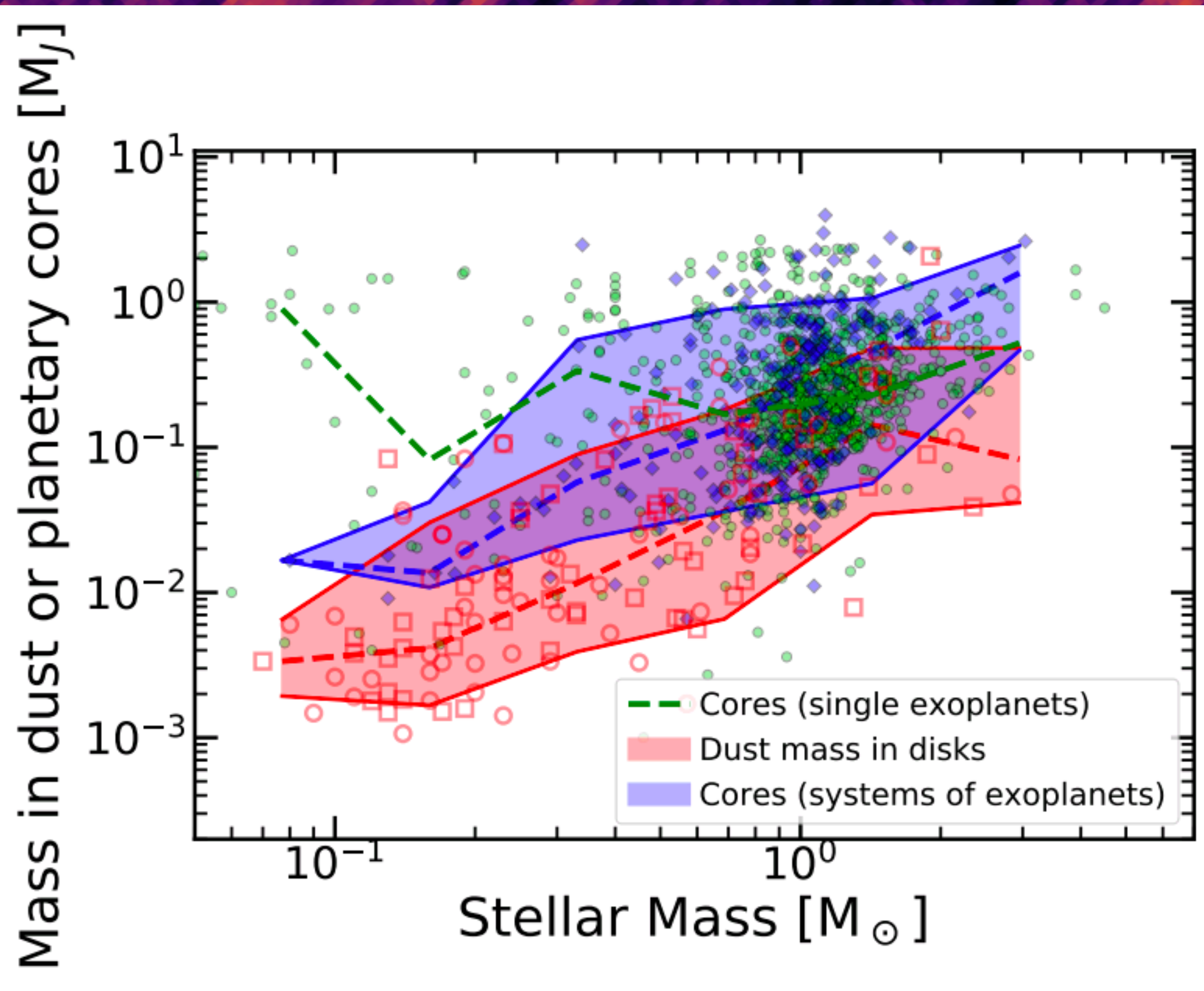
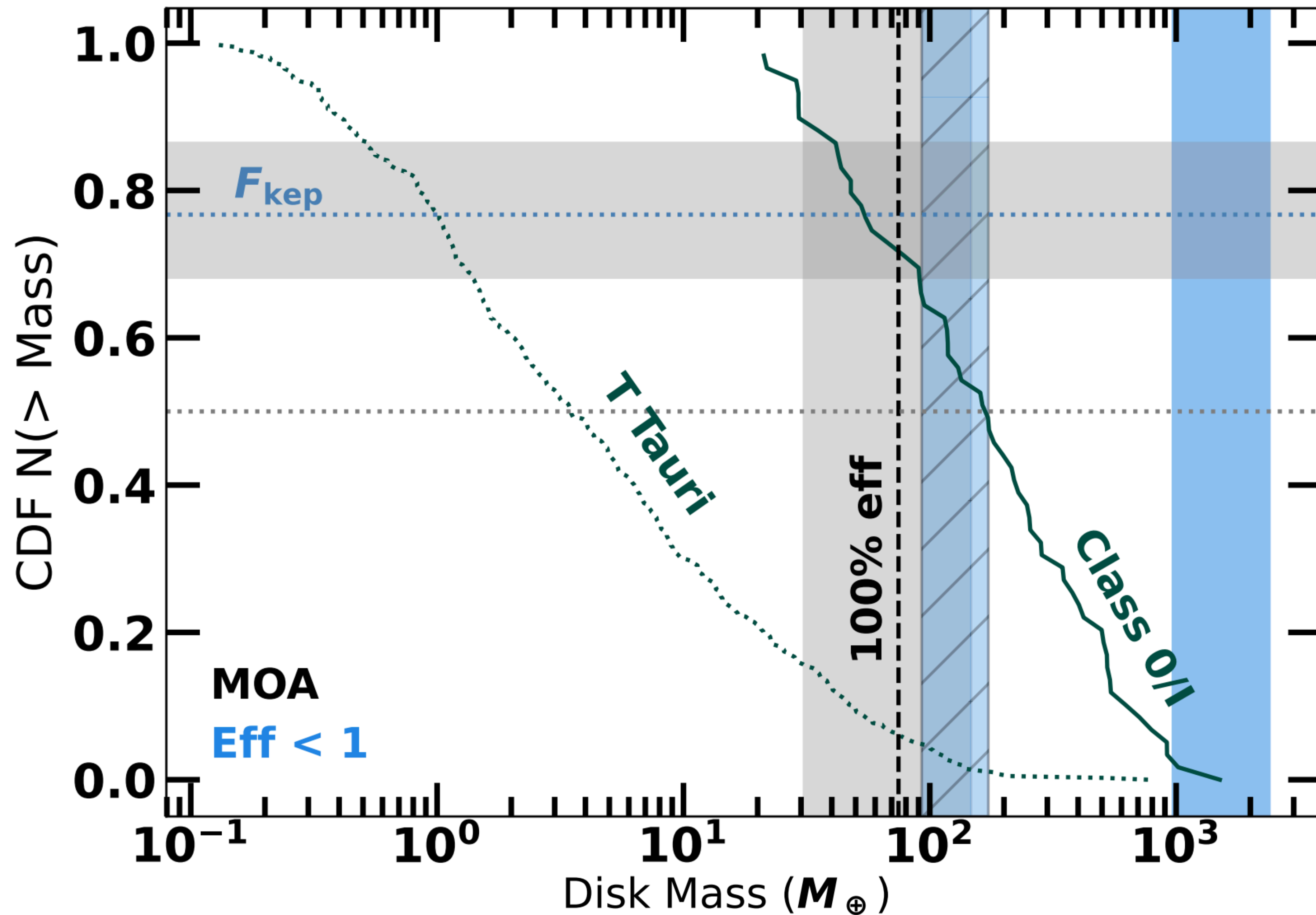


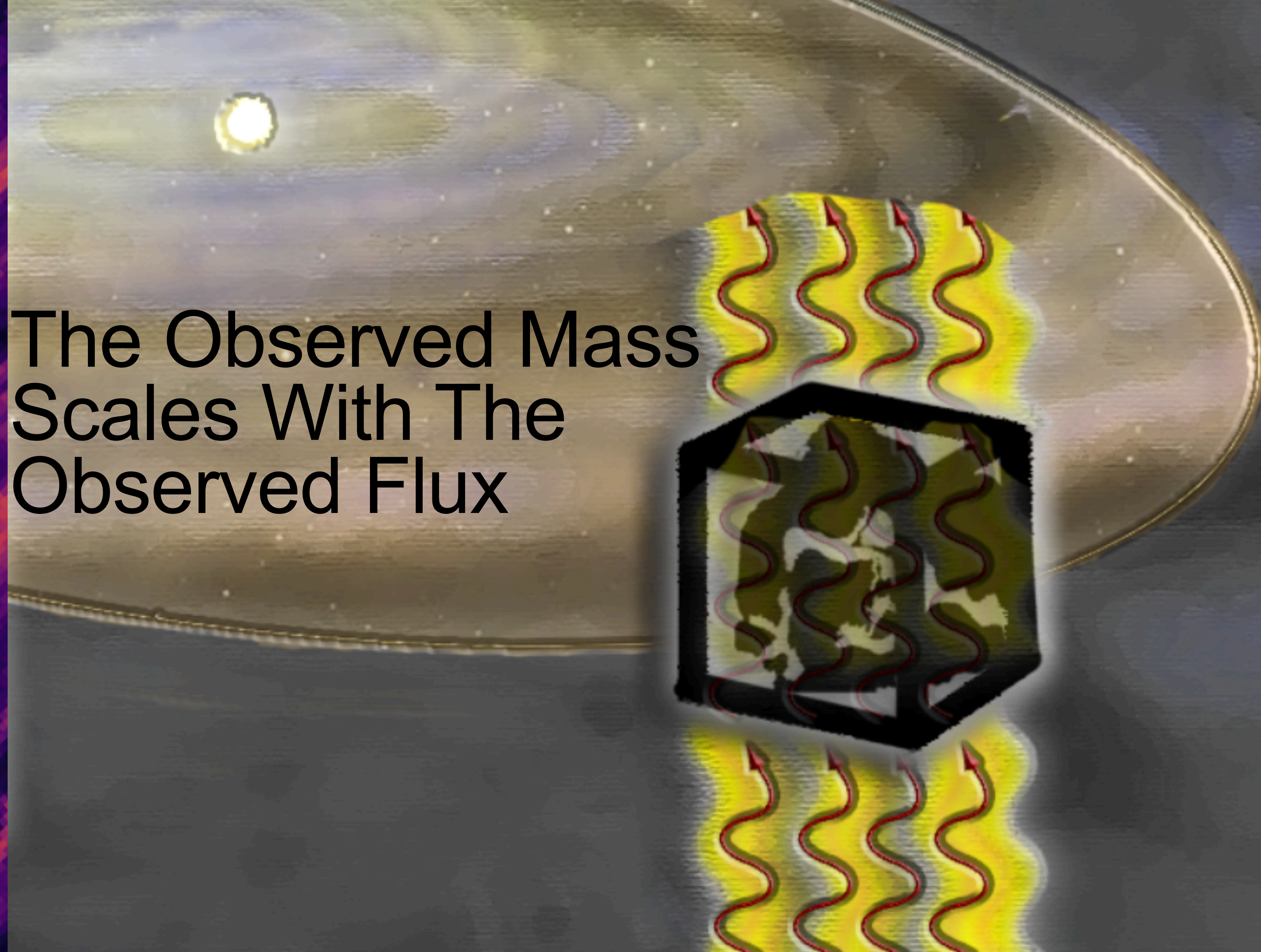
On The Mass Budget Problem: Streaming Instability and Optically Thick Emission



Pebble Accretion

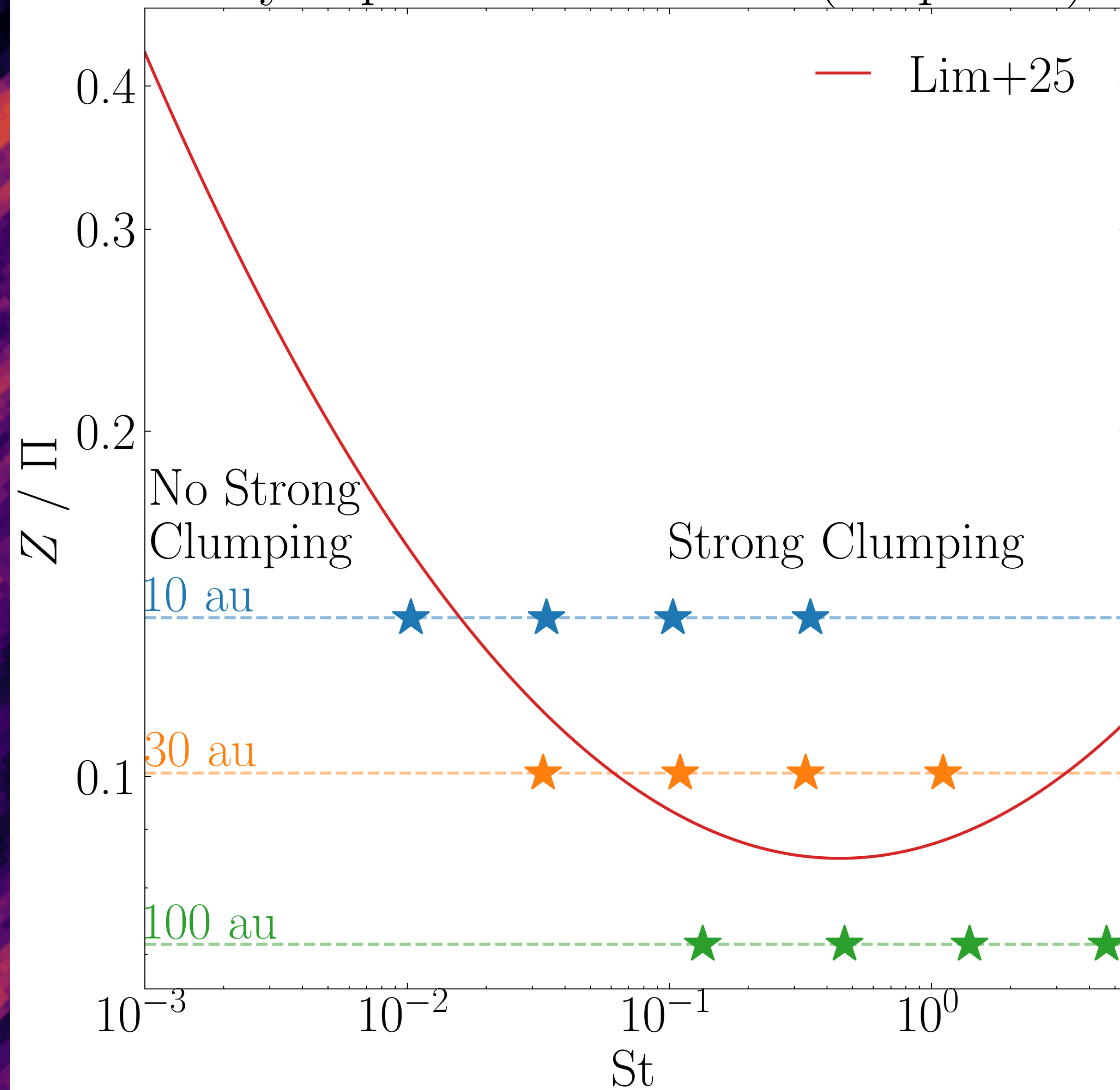


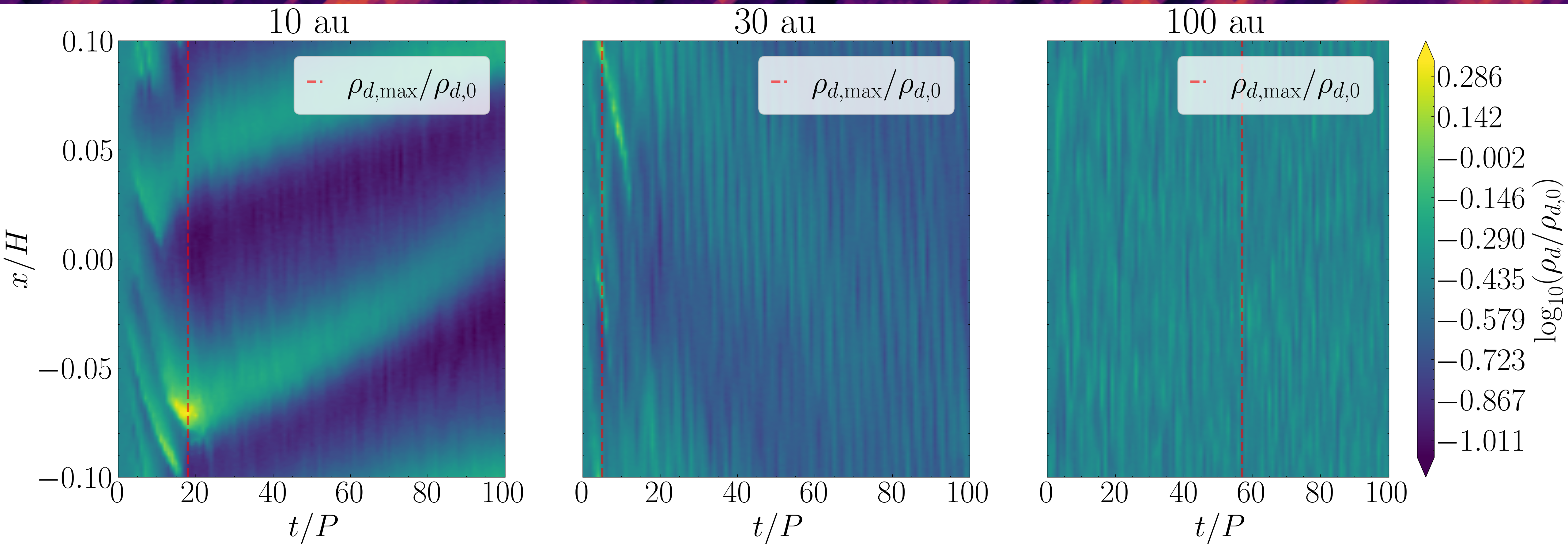
The Observed Mass Scales With The Observed Flux



Optically Thin Approximation (Hildebrand 1983)

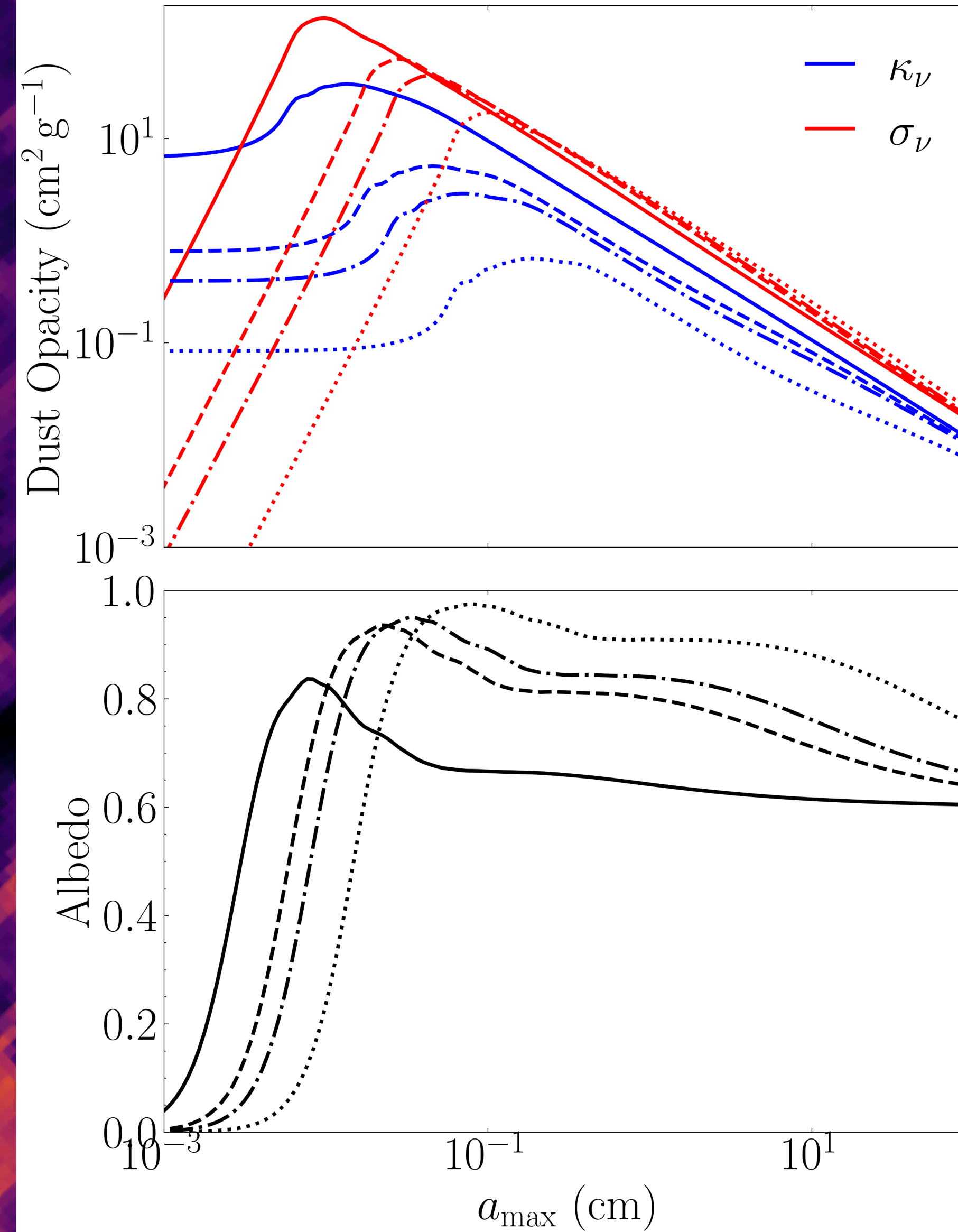
Streaming Instability Strong Clumping Regime
Polydisperse Simulations (4 Species)



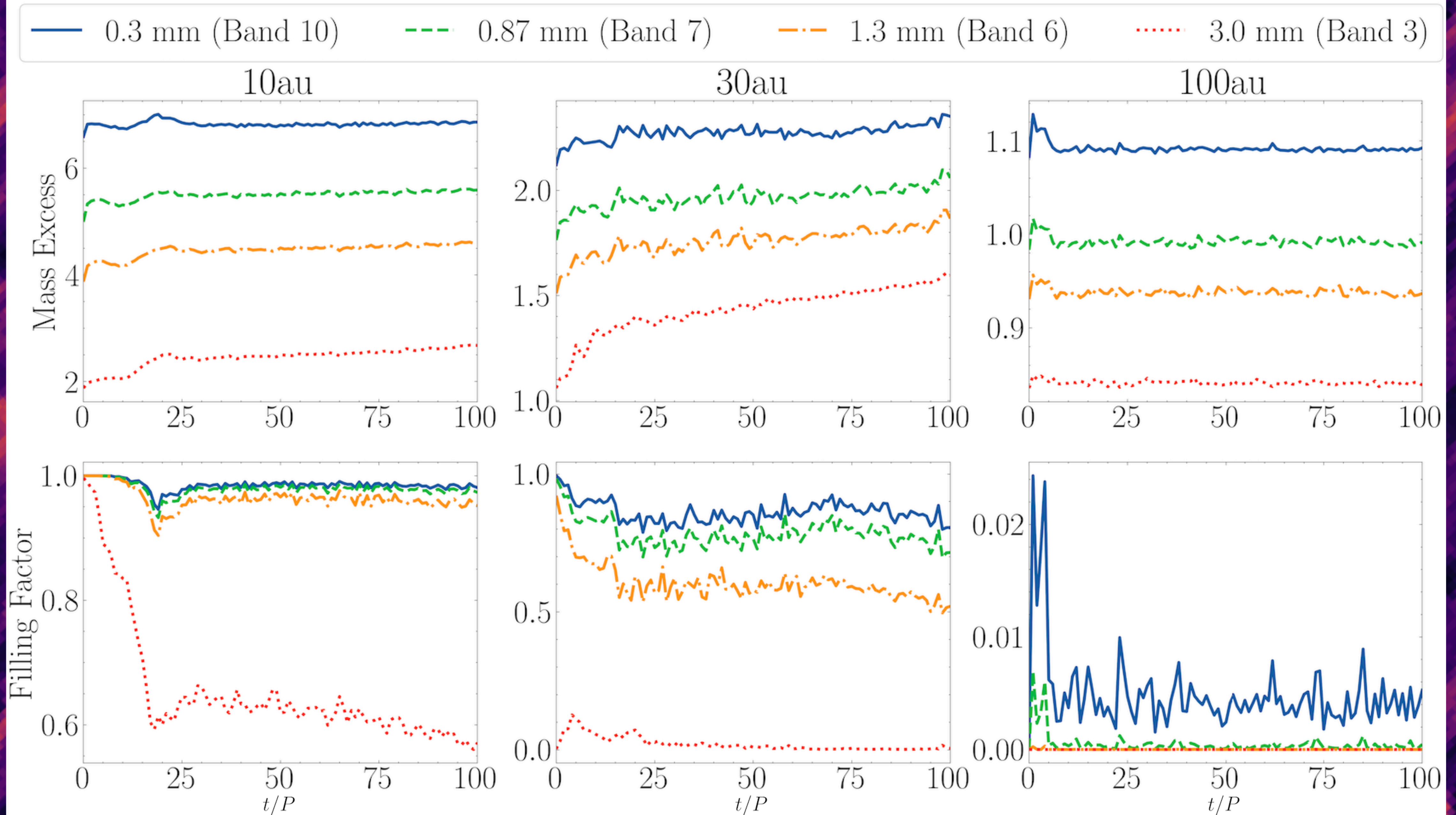


Size Averaged Opacities at ALMA Bands

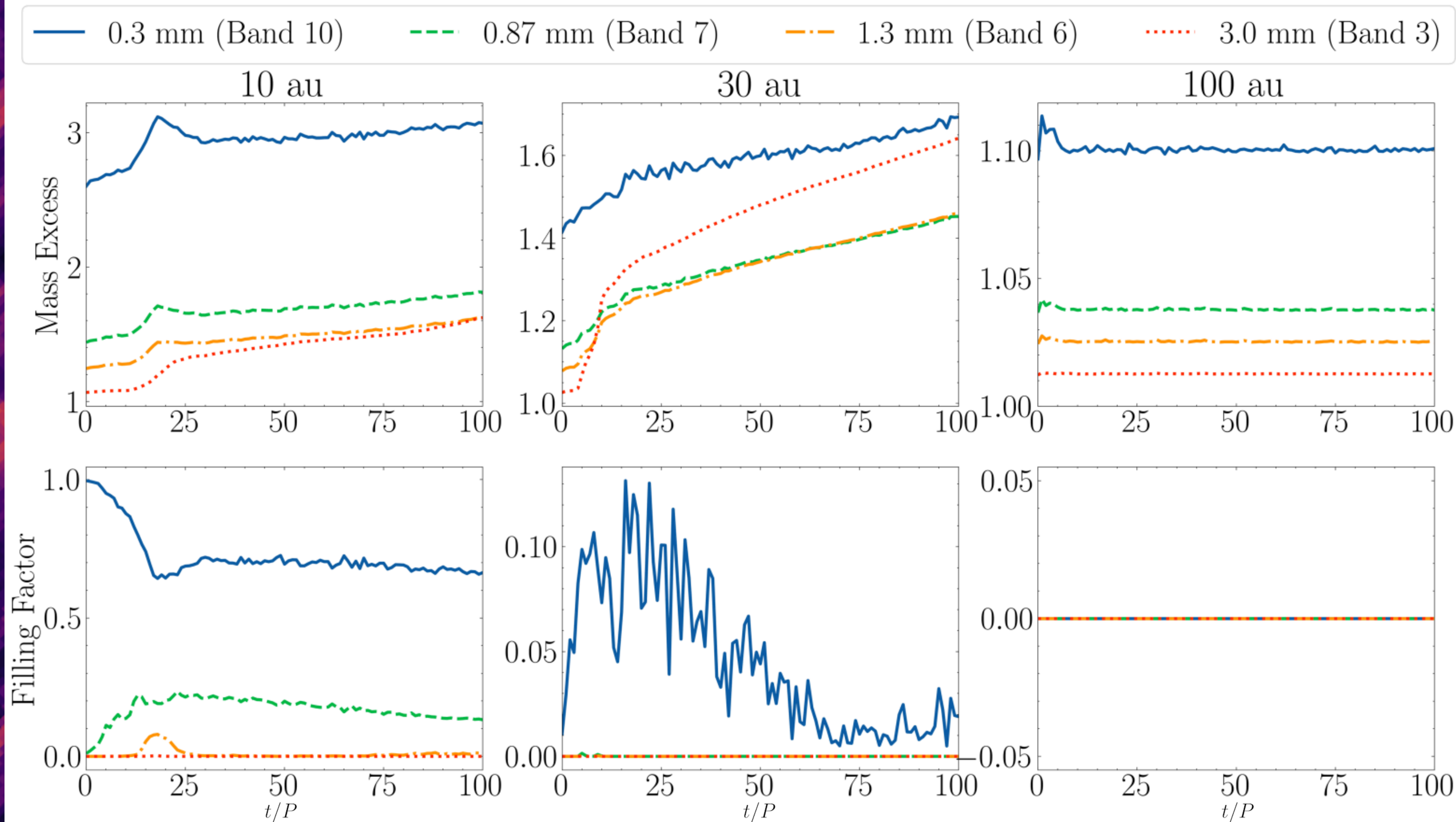
— 0.3 mm (Band 10) -·- 1.3 mm (Band 6)
 -- 0.87 mm (Band 7) 3.0 mm (Band 3)



Radiative Transfer Results

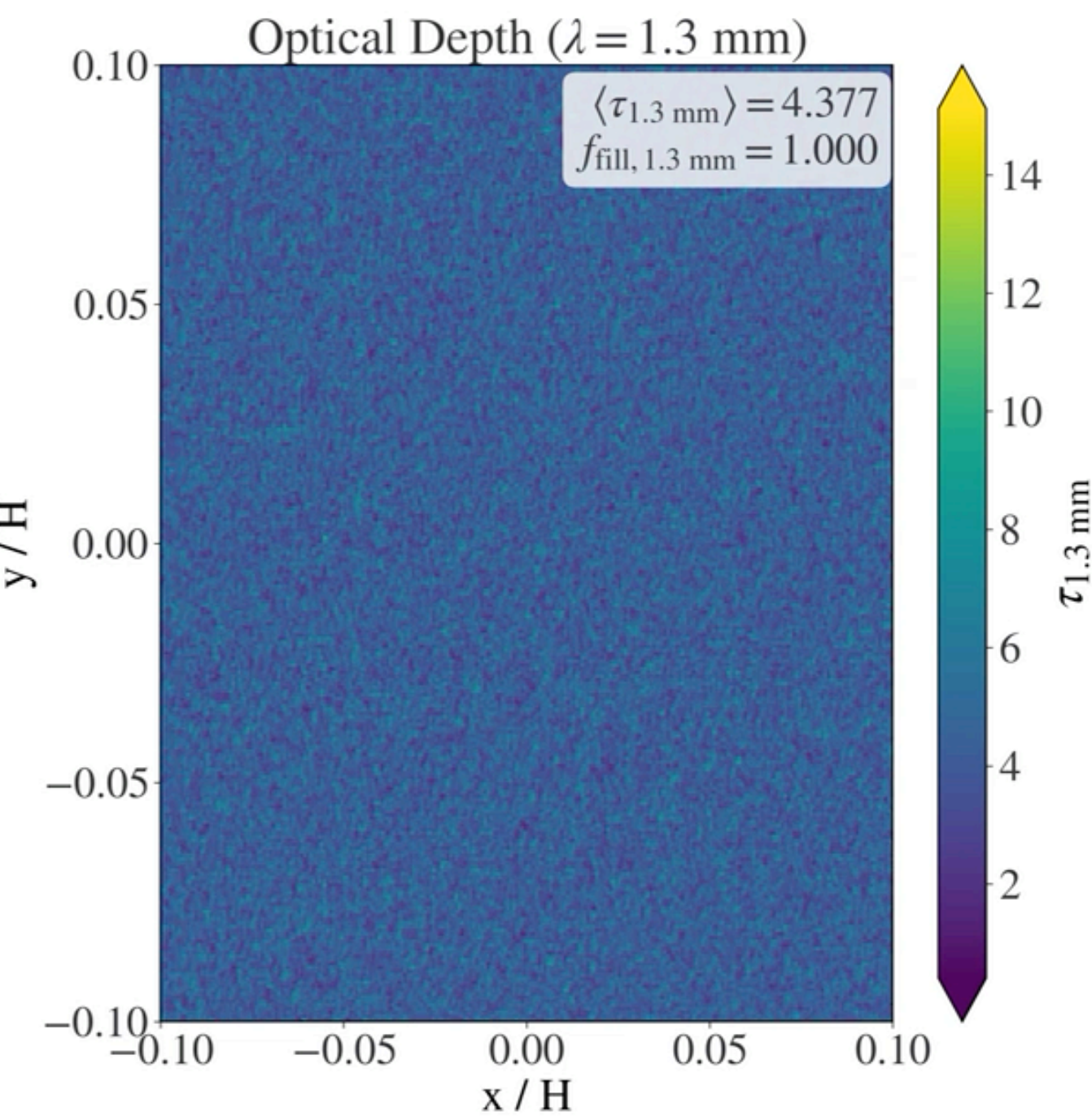
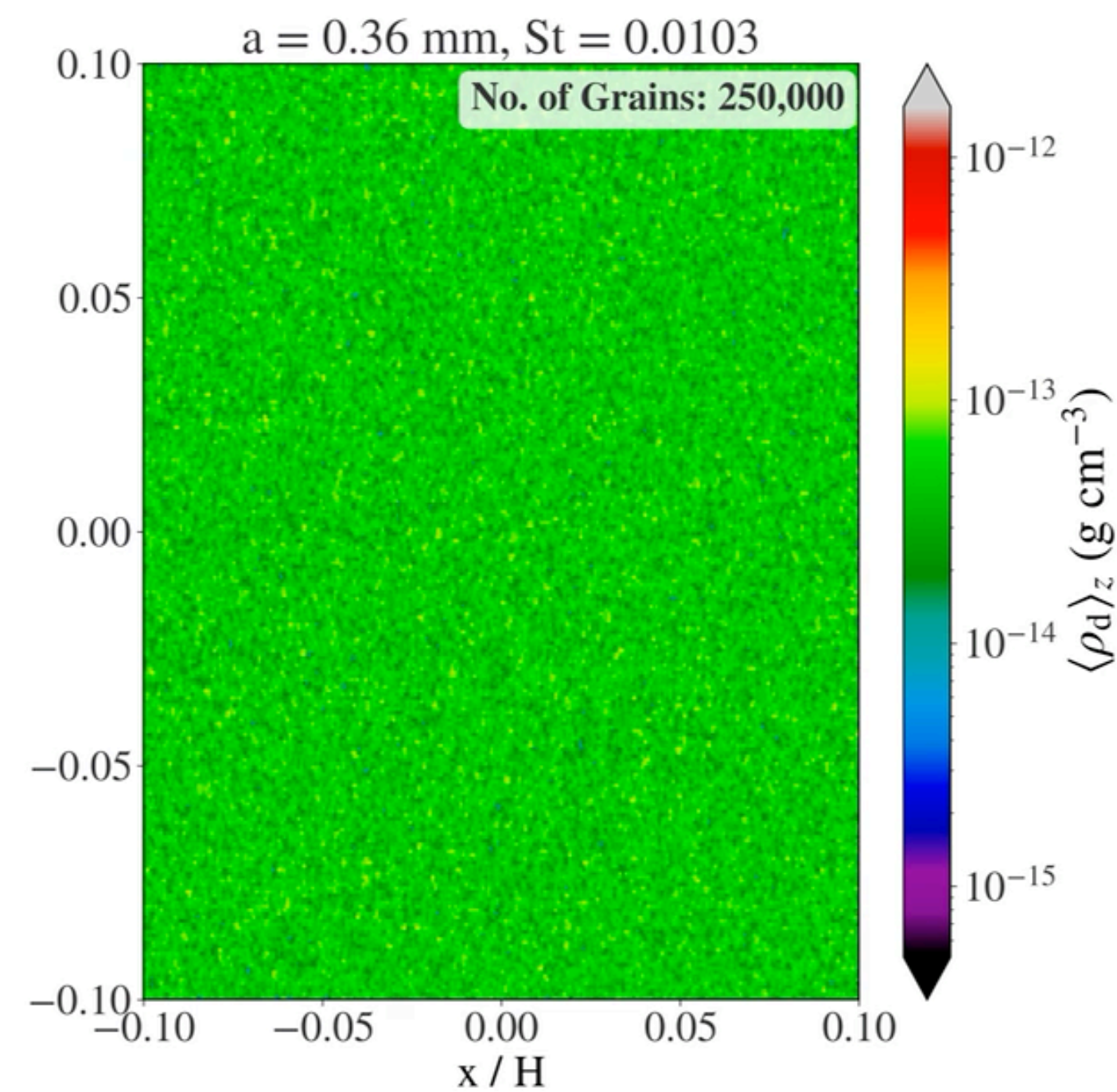
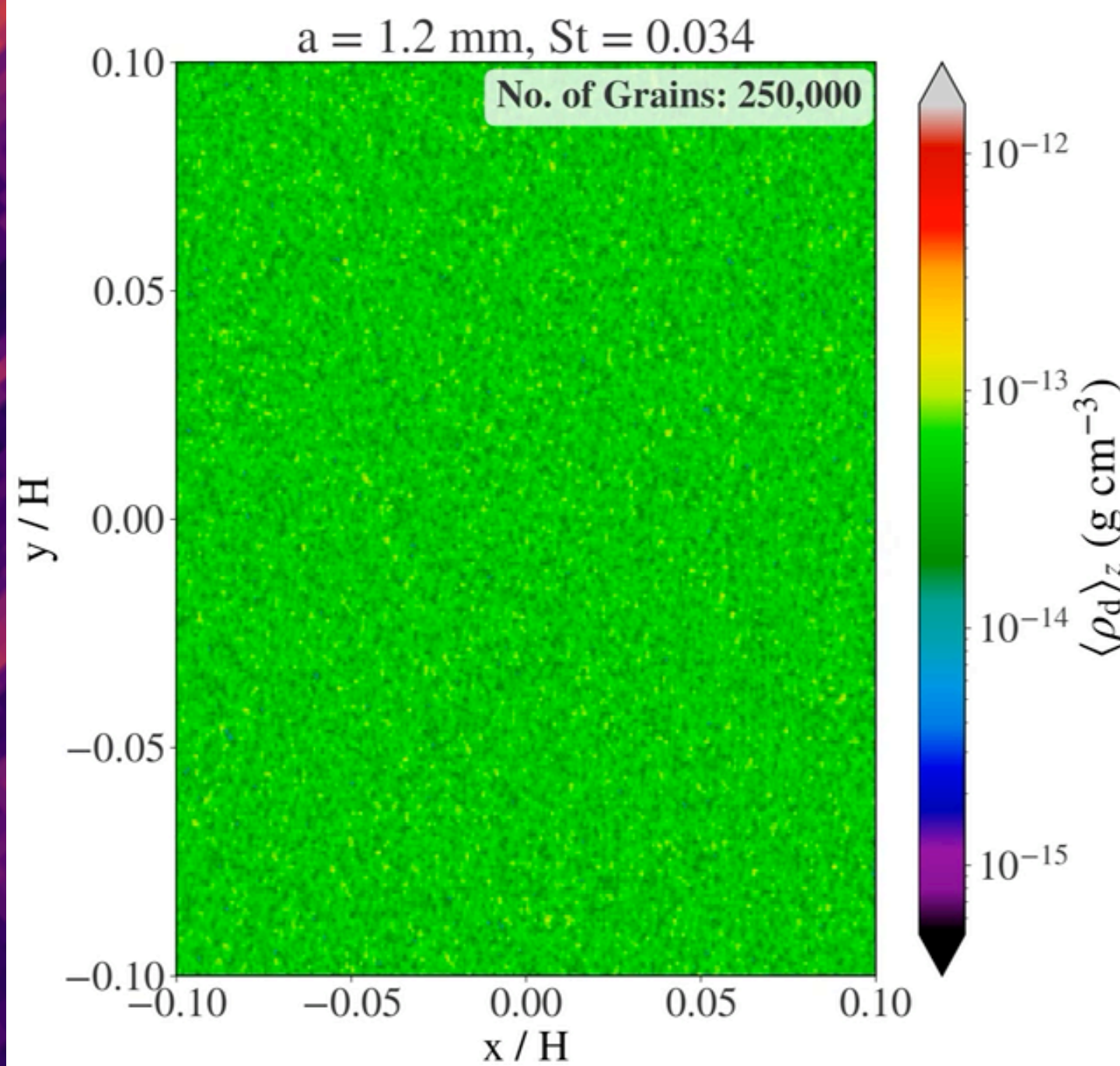
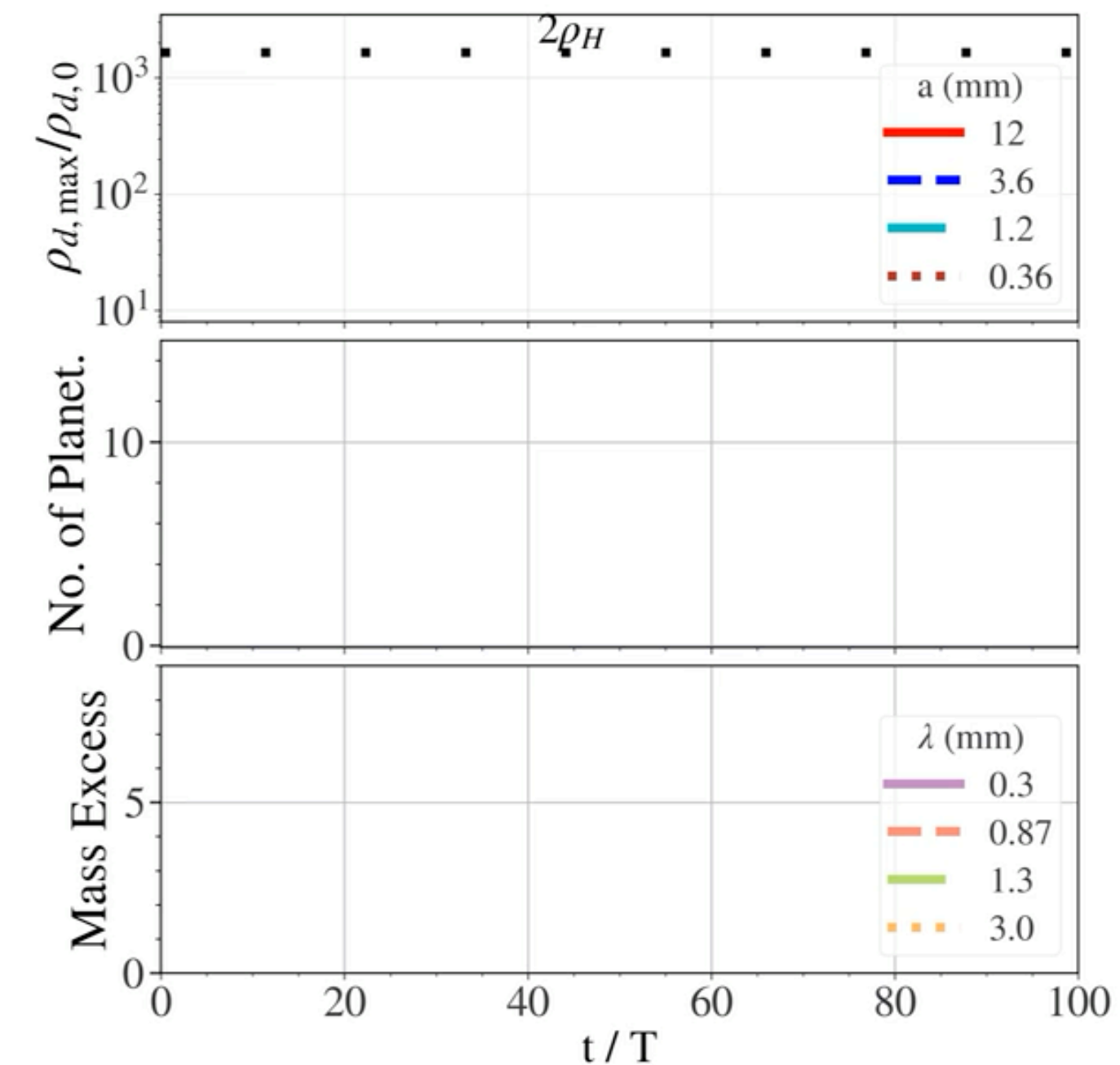
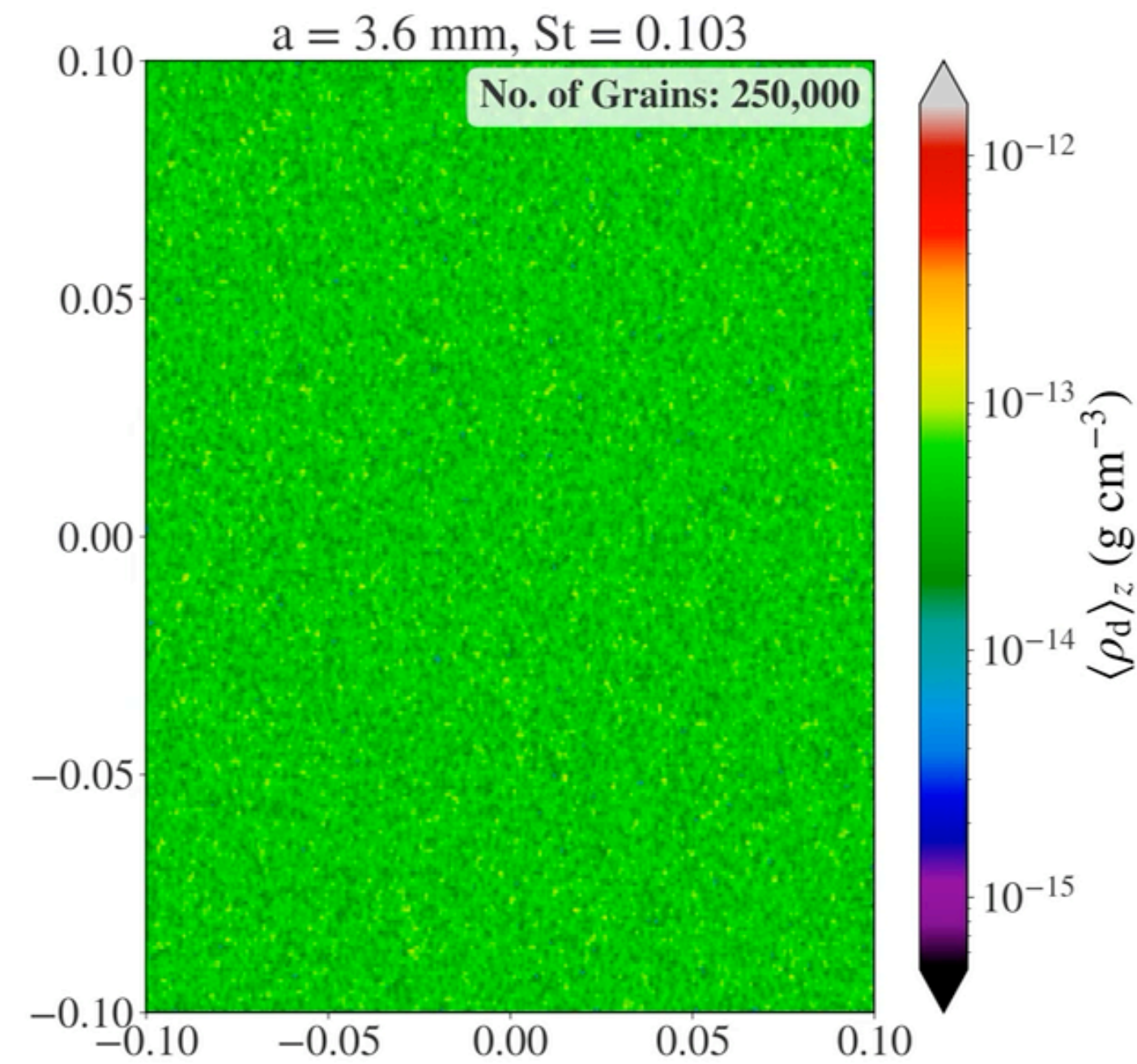
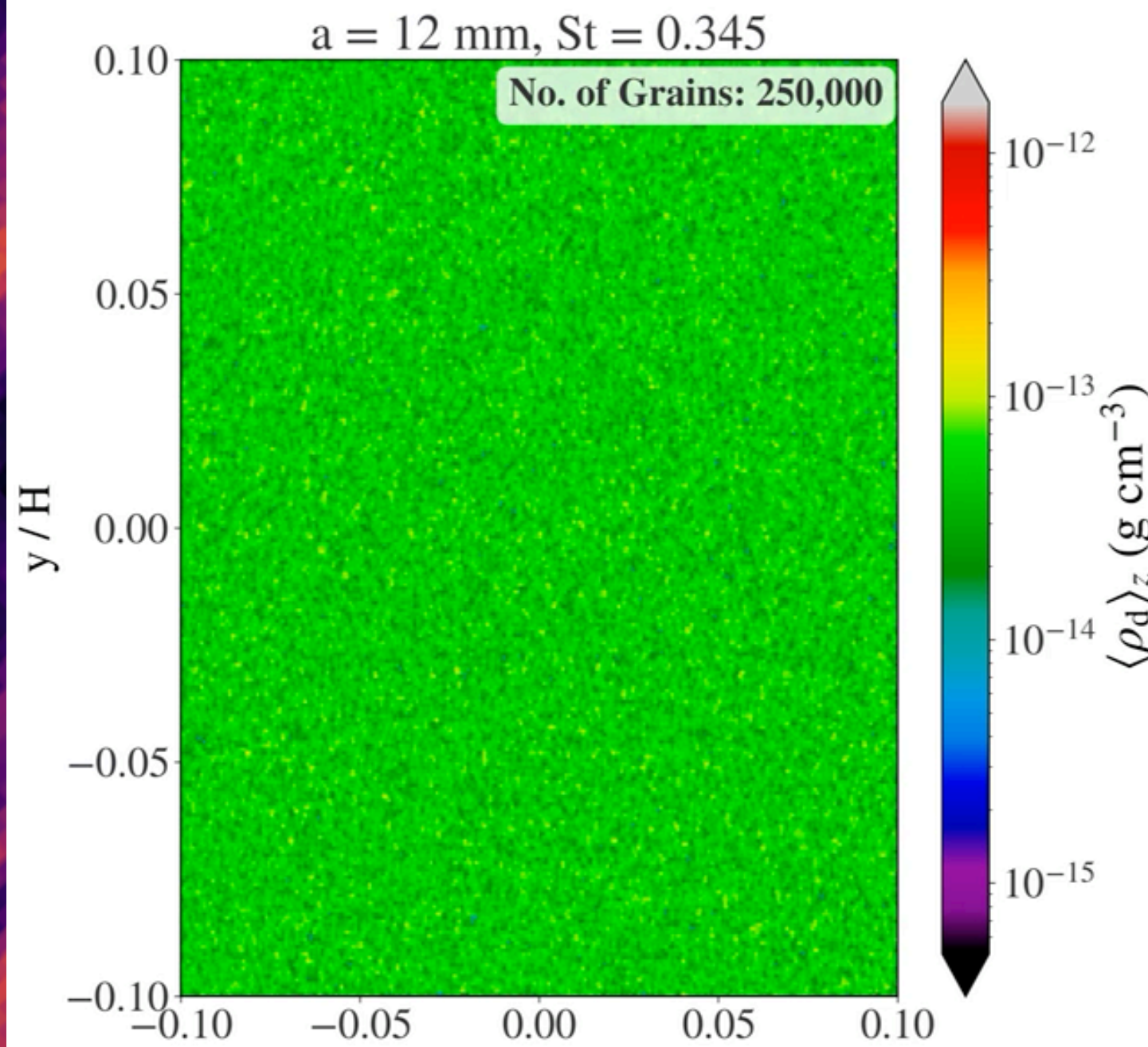


Radiative Transfer Results: Absorption-only



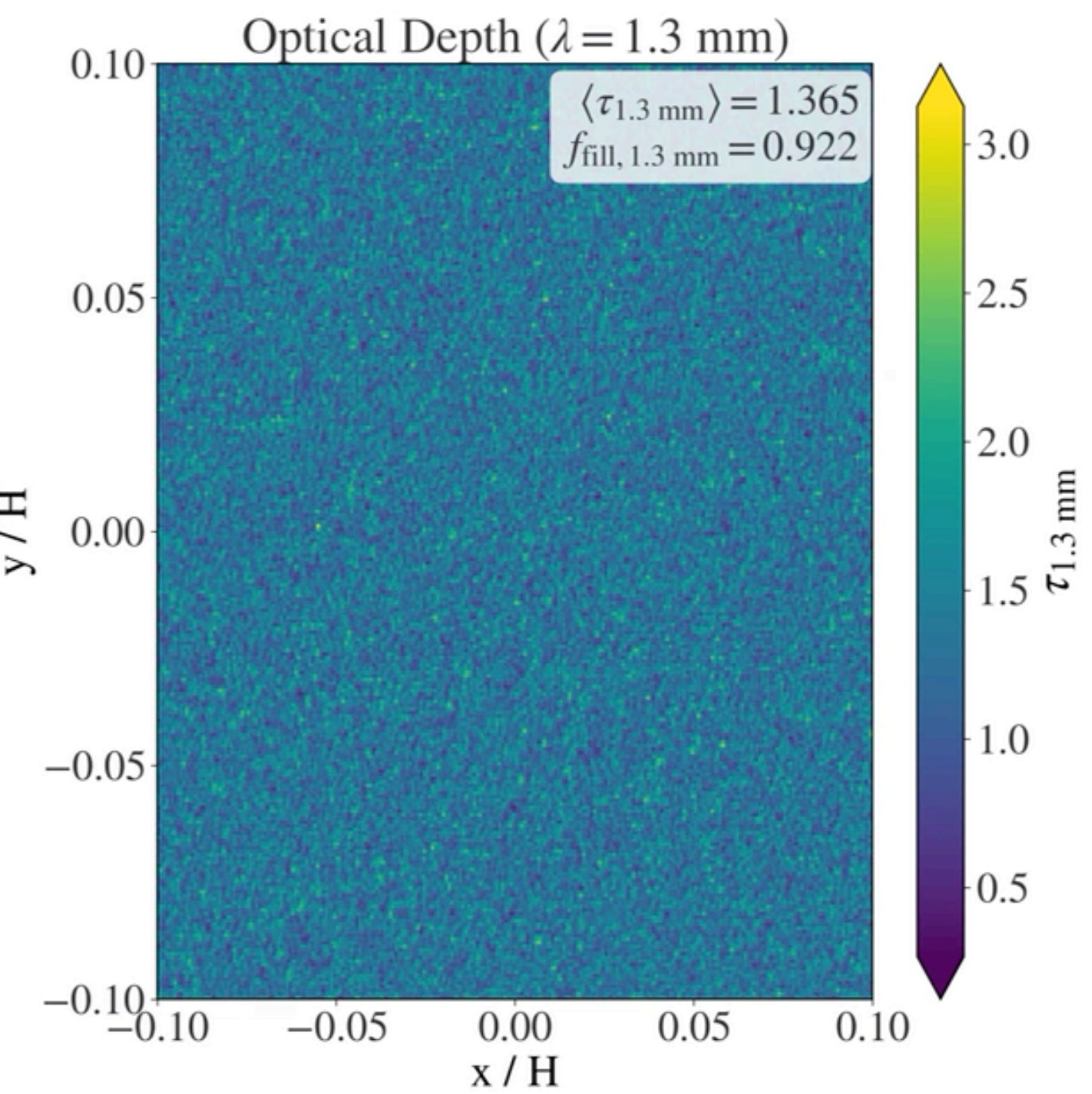
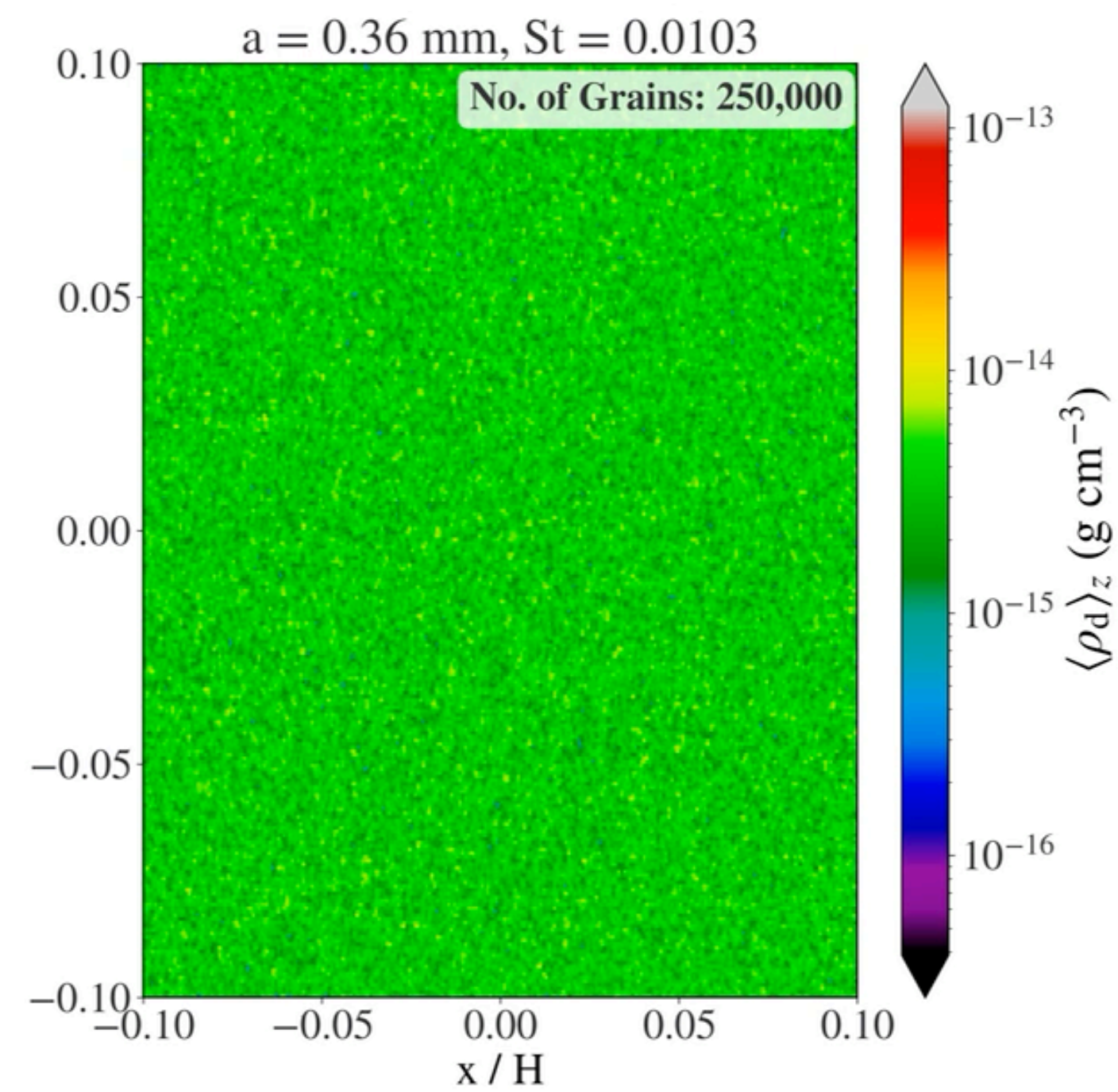
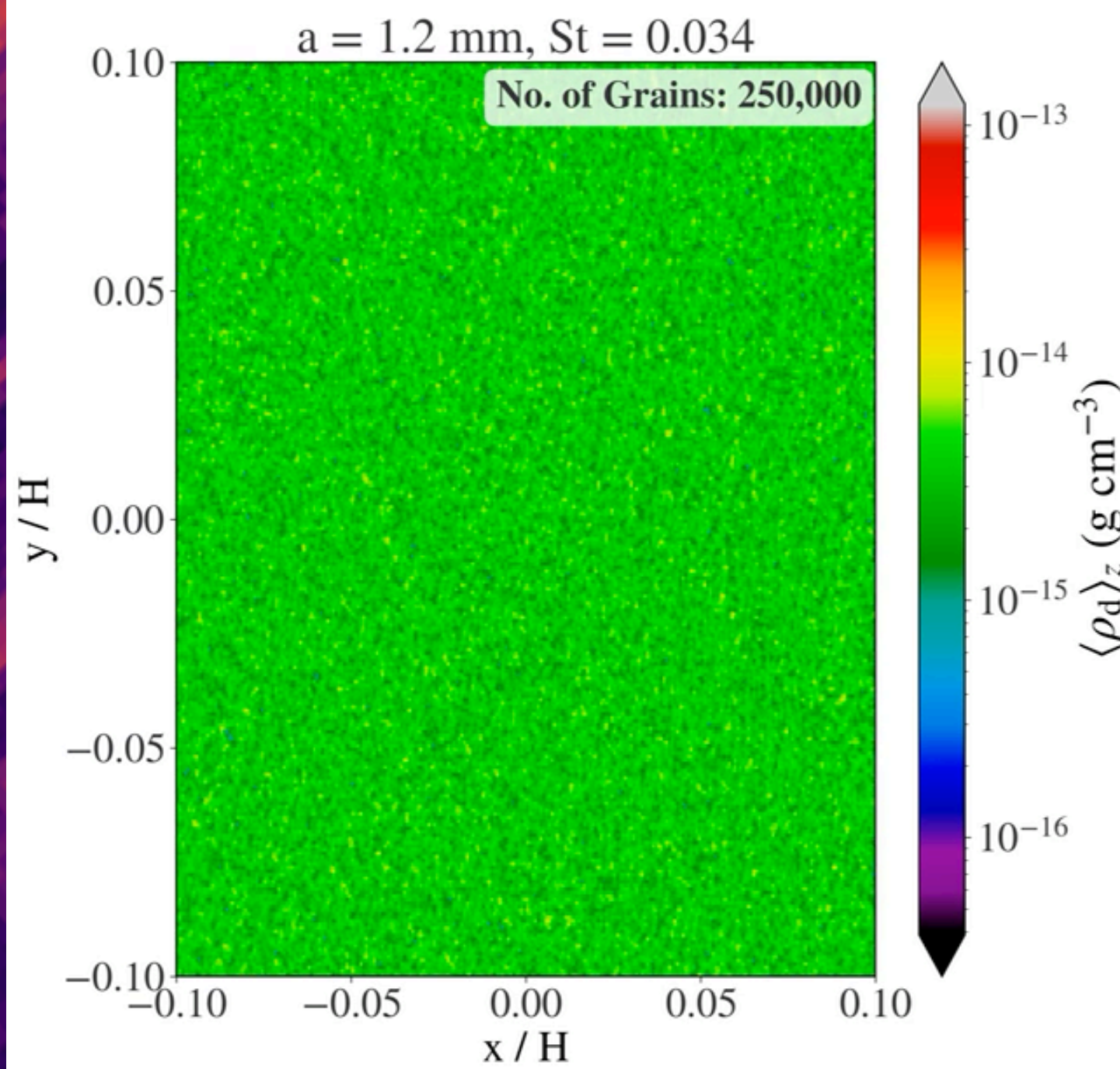
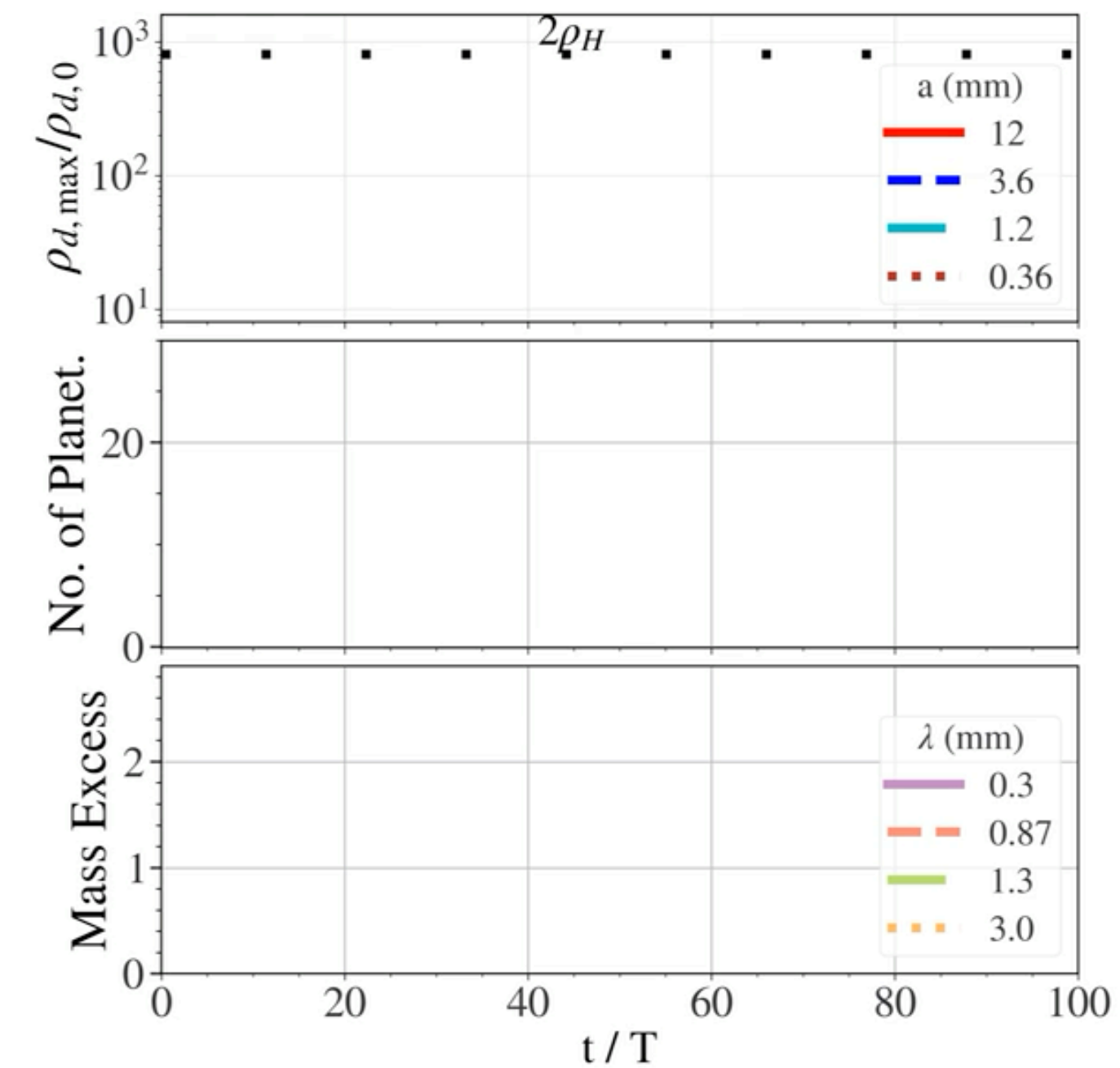
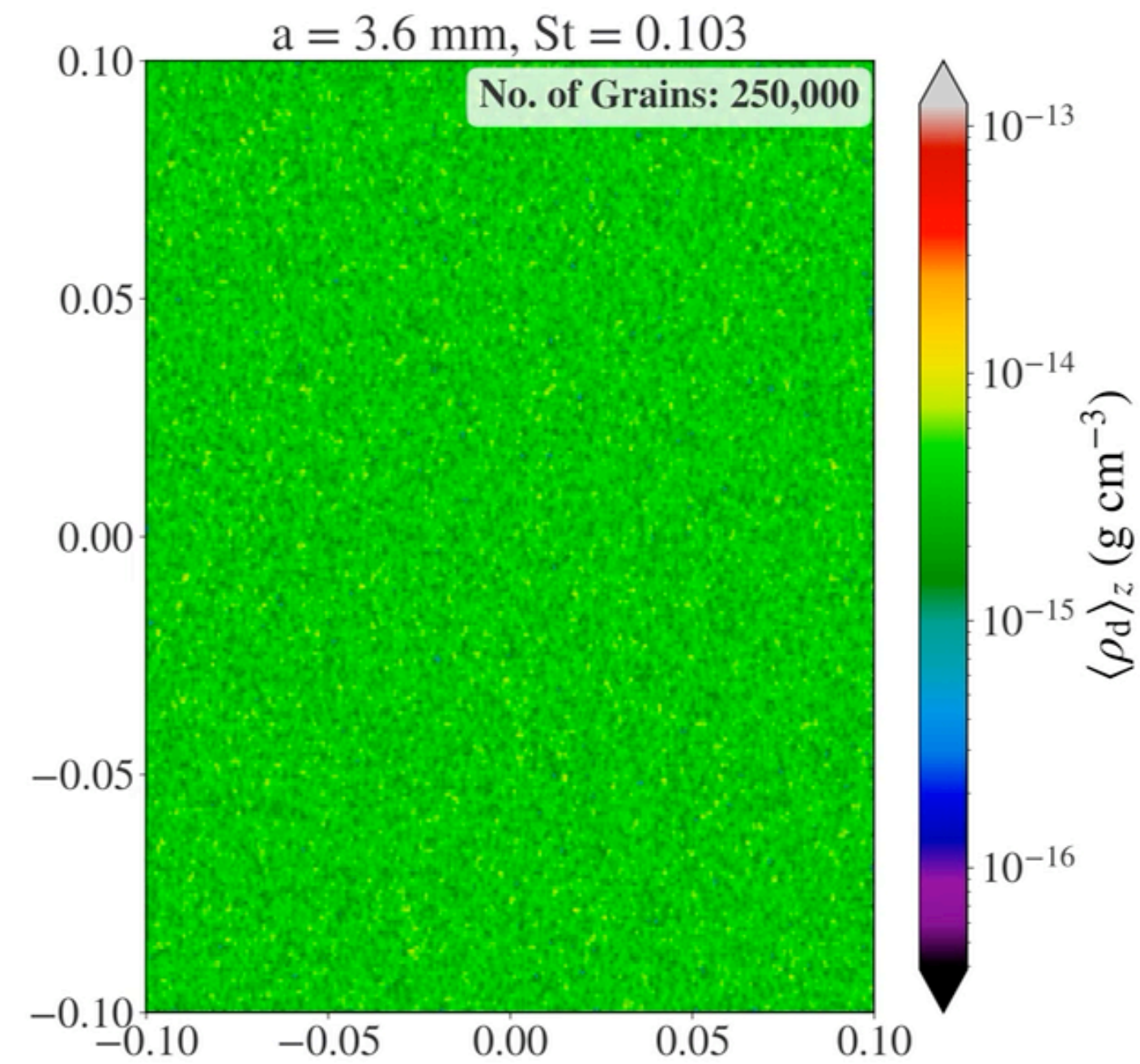
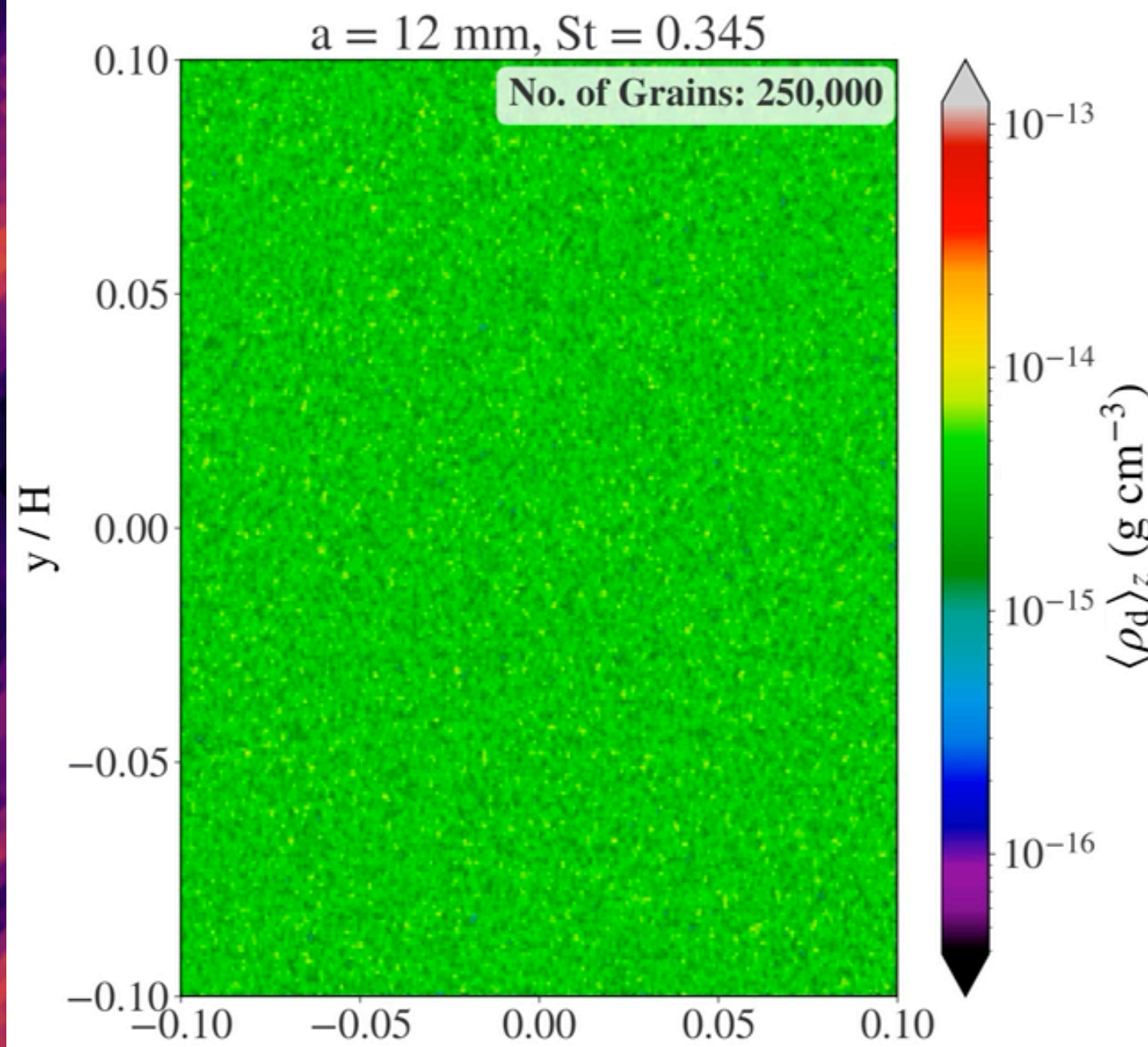
Self-Gravitating Polydisperse Streaming Instability ($r = 10$ au)

$t / T = 0$

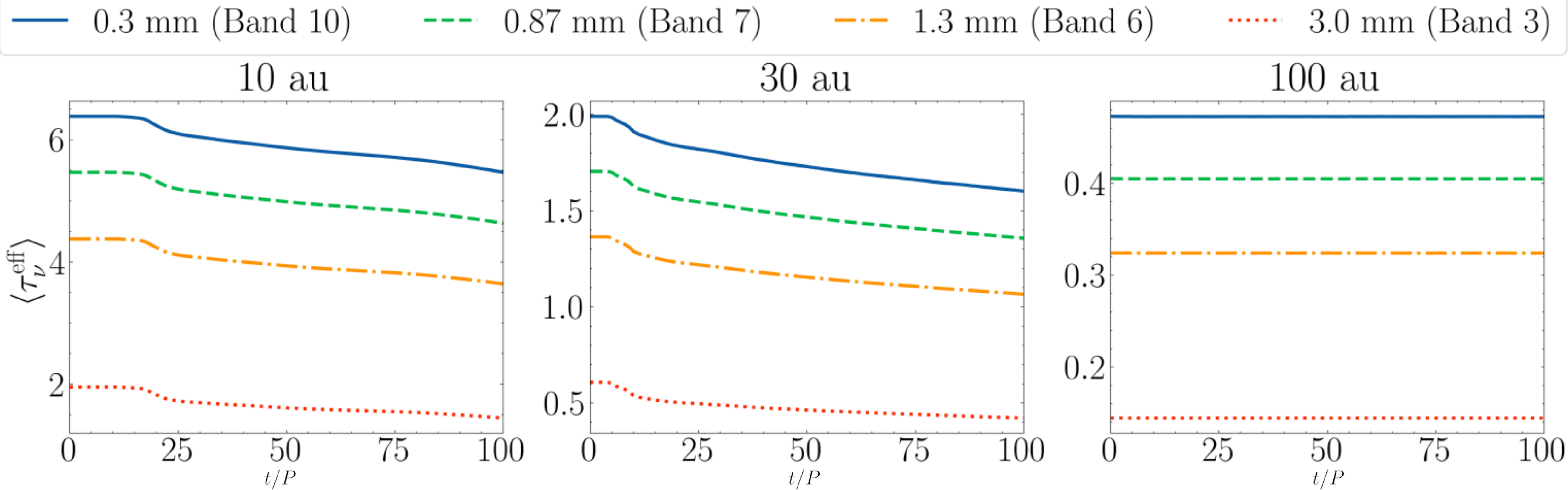


Self-Gravitating Polydisperse Streaming Instability ($r = 30$ au)

$t / T = 0$



Radiative Transfer Results





W. Lyra



L. Ricci



C.-C. Yang



J. B. Simon



J. Lim



D. Carrera

Thank You Co-Authors



D. Sengupta

Special Thank You



Questions?